

EVOLVEWEB AI – AI POWERED WEBSITE BUILDER

**A Project Report Submitted in Fulfillment of the
Requirements for the Award of Degree**

Of

**Bachelor of Technology In
Computer Science and Engineering**

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**Department of Computer Science & Engineering
M. M. Engineering College, Mullana, Ambala, Haryana, India**

CANDIDATE'S DECLARATION

We hereby certify that the work presented in the project entitled “EvolveWeb AI – AI Powered Website Builder” in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering of M.M. Engineering College, Mullana, Ambala, Haryana, India, is an authentic record of our own work carried out during the period from January 2026 to April 2026, under the supervision of Ms. Arpneek Kaur.

The matter presented in this project report has not been submitted by us for the award of any degree of this or any other Institute/University.

This is to certify that the above statement made by the above candidates is correct to the best of my knowledge.

Date: 07-04-2026

Signature of supervisor

Designation: Assistant Professor

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TABLE OF CONTENTS

Candidate's Declaration	ii
Acknowledgement	iii
Chapter 1. Introduction about Project	
1.1 Description about Work	1
1.2 Reasons for Selection of this Project Work	2
Chapter 2. Platform Requirements	3
Chapter 3. Objectives of Project Work	4
Chapter 4. Methodology Used	5
4.1 Proposed Algorithm or Technique	5
4.2 Related Diagrams (Use Case / ER / Activity / Sequence)	6
Chapter 5. Results & Discussion	7
5.1 Screenshots Obtained by Running Project	7
5.2 Description about Obtained Result	8
Chapter 6. Conclusion and Future Scope	9
REFERENCES	10

Chapter 1: Introduction about Project

1.1 Description about Work

EvolveWeb AI is a modern web-based application developed to automate the process of creating professional, responsive, and customizable websites. In today's digital era, having an online presence is essential for businesses, startups, students, and professionals. However, traditional website development requires technical knowledge of programming languages, frameworks, and deployment processes that many individuals simply do not possess.

This system uses artificial intelligence to automate the entire website creation process. Users are required to enter their business idea or project description in plain natural language. Based on this input, the system intelligently processes the idea through a five-stage AI pipeline and generates a fully structured, responsive, and production-ready website automatically.

The application is built using the MERN Stack (MongoDB, Express.js, React.js, Node.js) and integrates with the OpenRouter API using the DeepSeek language model. It includes secure JWT-based user authentication, a project management dashboard, real-time AI generation progress indicators, a live website preview system using a sandboxed iframe, and a ZIP export feature for downloading the generated website.

Overall, EvolveWeb AI reduces manual effort, saves significant time, and democratizes digital presence creation by making professional website building accessible to everyone, regardless of technical background.

1.2 Reasons for Selection of this Project Work

The main reason for selecting this project is the growing need for an accessible, automated, and intelligent website creation solution. Many students, startups, and small businesses lack the technical knowledge required to build professional websites, which often leads to missed digital opportunities despite having valuable ideas.

Another reason is the advancement of artificial intelligence and Large Language Models (LLMs) in modern applications. Integrating AI into website building through a multi-stage pipeline provides intelligent content generation, automated code production, and structured outputs that greatly exceed the capabilities of template-based builders such as Wix, Squarespace, and Webflow.

This project also provides practical exposure to real-world technologies such as React.js, Node.js, Express.js, MongoDB, JWT authentication, and AI API integration through OpenRouter. It provides a comprehensive understanding of full-stack development, REST API design, and AI pipeline engineering.

Furthermore, the project solves a genuine real-life problem — the inaccessibility of professional website creation for non-technical individuals — and delivers a working, deployable SaaS solution. Hence, it was selected as the final-year project for its practical value, technical depth, and innovative approach.

Chapter 2: Platform Requirements

The project requires both hardware and software components.

Hardware Requirements:

- Computer/Laptop with minimum 4GB RAM (8GB recommended for development)
- 2 GHz or faster processor
- Stable internet connection (required for AI API calls and MongoDB Atlas)
- Web Browser (Google Chrome / Microsoft Edge / Mozilla Firefox)

Software Requirements:

- Frontend: HTML, CSS, JavaScript, React.js (v18+), Tailwind CSS, Framer Motion, React Router DOM
- Backend: Node.js (v18+), Express.js, OpenAI SDK (via OpenRouter API)
- Database: MongoDB Atlas, Mongoose ODM
- Authentication: JSON Web Token (JWT), bcryptjs
- AI Service: OpenRouter API with DeepSeek model (deepseek/deepseek-chat)
- Export: Archiver library (ZIP generation), Blob URL (sandboxed iframe preview)
- Development Tools: Visual Studio Code, GitHub, Nodemon, Vite, Postman
- Browser: Chrome / Edge

Chapter 3: Objectives of Project Work

The primary objective of EvolveWeb AI is to develop an intelligent, AI-powered system that fully automates the website creation process from a plain-language idea description, eliminating the need for any programming or design knowledge.

An important objective is to implement a five-stage AI processing pipeline — covering idea analysis, idea evolution, blueprint generation, website structure planning, and final code generation — that produces structured, production-ready results at every stage.

The project also aims to provide a secure, scalable platform with JWT-based authentication, bcrypt password hashing, and MongoDB Atlas-backed project storage, ensuring both data security and system reliability.

Another objective is to deliver a user-friendly dashboard interface built with React.js, Tailwind CSS, and Framer Motion that provides real-time AI generation feedback, live website preview, and project management capabilities.

Additionally, the system enables quick generation, live preview, and ZIP download of generated websites, making it convenient for users to immediately deploy their AI-created web presence. Overall, the project aims to combine AI technology and full-stack development to create an efficient, intelligent, and accessible website-building solution.

Chapter 4: Methodology Used

4.1 Proposed Algorithm or Technique

The system follows a structured five-stage AI pipeline process:

1. User registers and logs into the system using JWT-authenticated credentials.
2. User enters a natural language description of their startup idea or business concept in the dashboard.
3. Stage 1 – Idea Analysis: AI scores the idea (0–100) and identifies strengths, weaknesses, and opportunities.
4. Stage 2 – Idea Evolution: AI refines and enhances the original concept into an improved, market-ready idea.
5. Stage 3 – Blueprint Generation: AI generates a comprehensive project blueprint including problem statement, target audience, core features, USP, monetization strategy, and future scope.
6. Stage 4 – Website Structure Planning: AI designs a multi-page website architecture with named sections per page.
7. Stage 5 – Code Generation: AI generates complete, responsive HTML/CSS/JavaScript website code based on the blueprint and structure.
8. The generated website is previewed live in a sandboxed iframe using Blob URLs and can be exported as a ZIP archive.

4.2 Related Diagrams (Use Case / ER / Activity / Sequence)

System Architecture

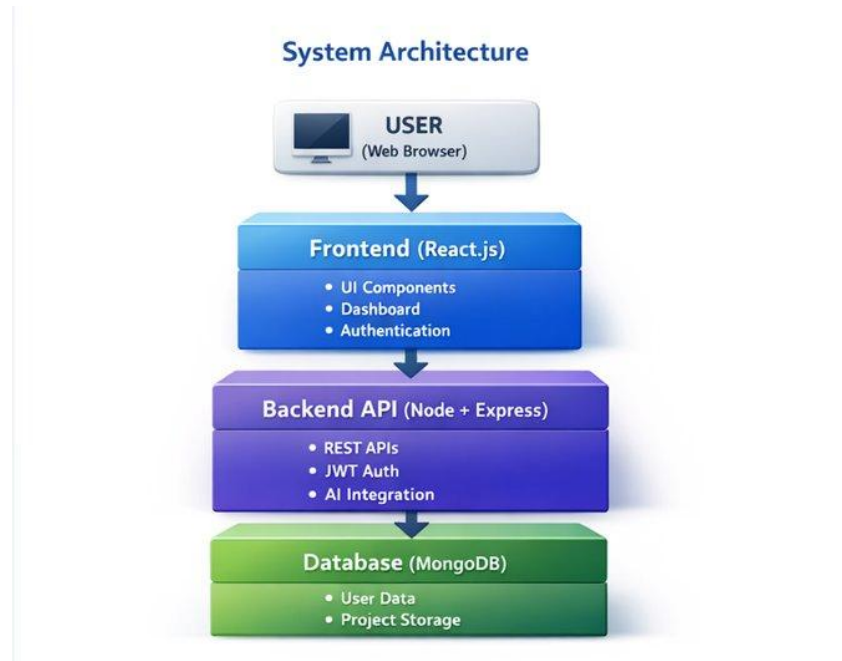


Figure 4.1: System Architecture of EvolveWeb AI

Data Flow Diagram – Level 0



Figure 4.2: Data Flow Diagram – Level 0 (Context Diagram)

ER Diagram

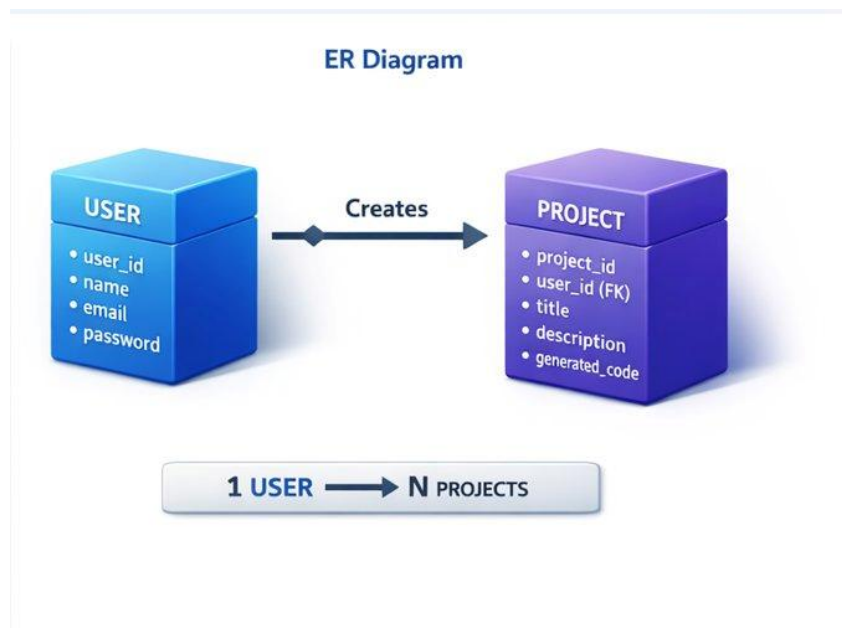


Figure 4.3: Entity Relationship Diagram

Project Timeline – Gantt Chart



Figure 4.4: Project Development Timeline (Gantt Chart)

Chapter 5: Results & Discussion

5.1 Screenshots obtained by running project

[Screenshot 1: Landing Page]

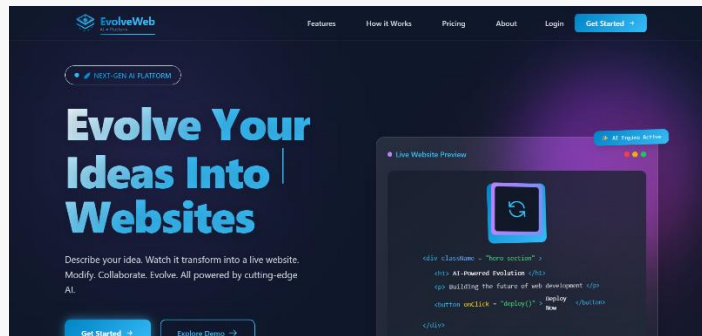


Figure 5.1: EvolveWeb AI – Landing Page

[Screenshot 2: Dashboard Home – AI Generation Interface]

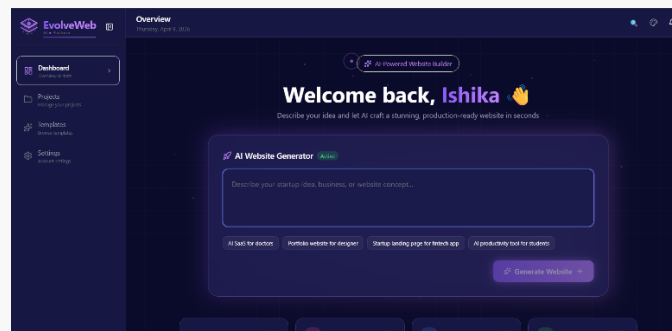


Figure 5.2: Dashboard Home – Project Creation & AI Generation

[Screenshot 3: Project Details – AI Analysis & Blueprint]

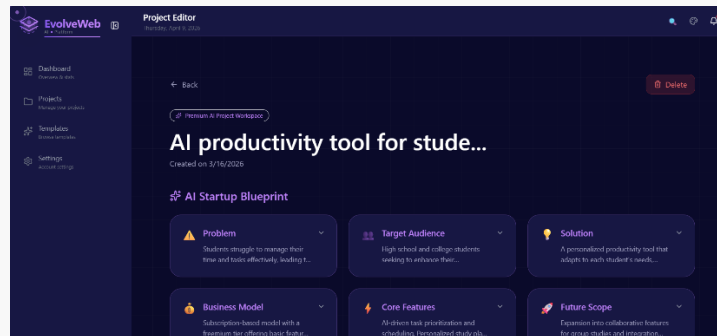


Figure 5.3: Project Details – AI Analysis and Website Blueprint

[Screenshot 4: Live Website Preview]

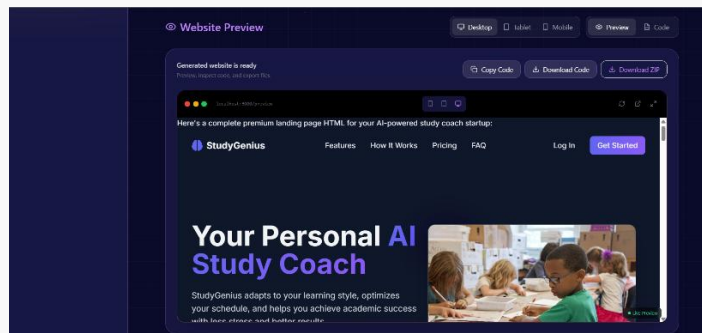


Figure 5.4: Live Website Preview with AI-Generated Code

5.2 Description about obtained result

EvolveWeb AI successfully generates professional, responsive websites based on user-provided startup idea descriptions. The system is efficient and produces structured, production-ready website code through its five-stage AI pipeline. The AI-generated content — including idea analysis scores, SWOT insights, business blueprints, and complete HTML/CSS/JavaScript code — is consistently high quality and immediately usable.

The application performs well in terms of speed and usability. Users can navigate through the system, trigger the full AI pipeline, and receive a live website preview within minutes. The JWT-based authentication works correctly, and all project data including AI-generated code is securely stored in MongoDB Atlas.

The live website preview renders correctly in the sandboxed iframe using Blob URLs, with no security issues. The ZIP download feature produces valid archives of the generated website files. Overall, the results confirm that the system meets all its defined objectives and delivers a genuinely useful, innovative solution for automated website creation.

Chapter 6: Conclusion and Future Scope

Conclusion

EvolveWeb AI is a useful, efficient, and innovative tool that fully automates the process of creating professional websites. It reduces manual effort and provides high-quality, production-ready output using a multi-stage artificial intelligence pipeline. The system is user-friendly, scalable, and suitable for students, startups, small businesses, and any non-technical individual who needs a digital presence.

The platform successfully demonstrates that modern Large Language Models can be integrated into a full-stack MERN application to deliver practical, real-world value. All defined objectives were achieved: automated website generation, secure authentication, project storage, live preview, and ZIP export. The project makes professional website creation accessible to everyone, regardless of technical background or financial resources.

Future Scope

The project can be further improved by adding advanced features such as one-click deployment integration with hosting platforms like Vercel and Netlify, allowing users to go live instantly after generation.

A drag-and-drop visual customization builder can be added to allow users to edit AI-generated layouts without writing code. AI-based SEO optimization can be integrated to improve the search engine ranking of generated websites.

The system can also support multiple languages to reach a wider global audience. A mobile application version can be developed for better accessibility. A template marketplace and SaaS billing model with Stripe integration can be implemented to monetize the platform commercially.

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